ENGINEERING MANAGEMENT

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Unit IV: Human Resource Management and Control (8 Hrs.)

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Specific Objectives

To enhance students' knowledge of human resource management and control functions, emphasizing their practical application for managing ICT organization

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4.1 Human Resource Management

Human resource management (HRM) is the practice of recruiting, hiring, deploying and managing an organization's employees. HRM is often referred to simply as human resources (HR). A company or organization's HR department is usually responsible for creating, putting into effect and overseeing policies governing workers and the relationship of the organization with its employees.

Human Resource Management (HRM) in the context of ICT-based projects and organizations refers to the practice of managing an organization's workforce using information and communication technology (ICT) tools to streamline processes like recruitment, onboarding, performance evaluation, training, and employee records, ultimately aiming to optimize workforce productivity and efficiency within a digital environment.

4.1.1 Functions of human resource management

Human Resource Management (HRM) plays a vital role in organizations across industries, serving as the linchpin between an organization's objectives and its workforce. The importance of HRM in organizations cannot be overstated, as it encompasses a wide array of functions essential for managing people effectively and driving organizational success. Below are some key points functions of HRM

Talent Acquisition and Recruitment

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- HRM plays a key role in talent acquisition. It is responsible for attracting, sourcing, and recruiting top talent to fill key roles within the organization.
- Effective recruitment strategies ensure that the organization has the right people with the necessary skills and qualifications to drive business growth and innovation.
- By identifying and selecting the best candidates, HRM contributes to building a skilled and diverse workforce that can meet the organization's current and future needs.

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Employee Development and Training

- HRM is tasked with providing employees with the training, development, and resources they need to excel in their roles and advance their careers.
- Training programs enhance employees' skills, knowledge, and competencies, enabling them to perform their jobs more effectively and contribute to the organization's success.
- Employee development initiatives also foster employee engagement, motivation, and loyalty, leading to higher levels of job satisfaction and retention.

Performance Management and Appraisal

- HRM establishes performance management systems to set clear performance expectations, provide feedback, and evaluate employee performance.
- Performance appraisals enable managers to assess employees' strengths and areas for improvement, identify training and development needs, and reward high performers.
- By aligning individual performance with organizational goals, HRM ensures that employees' efforts contribute to achieving the organization's objectives.

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Employee Relations and Engagement

- HRM plays a crucial role in fostering positive employee relations and creating a supportive work environment.
- HR professionals address employee concerns, grievances, and conflicts, promoting open communication and resolving issues in a fair and timely manner.

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- Employee engagement initiatives, such as recognition programs, wellness initiatives, and work-life balance initiatives, enhance morale, motivation, and productivity, leading to higher levels of employee satisfaction and retention.

Compliance with Employment Laws and Regulations

- HRM ensures that the organization complies with all relevant employment laws, regulations, and industry standards.
- HR professionals stay abreast of changes in labor laws and regulations, update policies and procedures accordingly, and advise management on legal compliance issues.
- By mitigating legal risks and ensuring fair and equitable treatment of employees, HRM helps protect the organization from potential lawsuits and reputational damage.

Organizational Development and Change Management

- HRM plays a critical role in driving organizational development initiatives and managing change effectively.
- HR professionals lead change management efforts, communicate organizational changes to employees, and help employees adapt to new ways of working.
- By promoting a culture of continuous improvement and innovation, HRM helps organizations remain agile, competitive, and resilient in the face of evolving business challenges.

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Key HR functions in an ICT environment:

- **Recruitment and Selection:**
  - Posting job openings on online job boards
  - Utilizing Applicant Tracking Systems (ATS) to screen candidates
  - Conducting virtual interviews
  - Assessing technical skills through online tests
- **Onboarding and Orientation:**
  - Digital onboarding modules
  - Remote introductions to company culture and systems
- **Performance Management:**
  - Real-time performance tracking through project management tools
  - Continuous feedback mechanisms via online platforms
  - Setting measurable goals aligned with organizational objectives
- **Learning and Development:**
  - Online training courses and e-learning platforms
  - Skill development programs tailored to emerging technologies
  - Personalized learning paths based on employee needs
- **Compensation and Benefits:**
  - Automated payroll processing
  - Digital benefits enrollment platforms
  - Performance-based incentives tied to digital metrics
- **Employee Relations:**
  - Virtual communication channels for employee feedback
  - Addressing employee concerns through online platforms
  - Maintaining employee engagement through virtual events
- **HR Analytics and Data Management:**
  - Utilizing data from HR systems to identify trends and patterns
  - Data-driven decision making to optimize workforce strategies

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- Ensuring data privacy and security compliance

Important considerations for HR in ICT environments:

- **Adapting to rapid technological change:** Stay updated with emerging technologies and their impact on the workforce.
- **Building a strong virtual culture:** Foster collaboration and engagement in a remote working environment.
- **Cybersecurity awareness:** Educate employees about data protection and privacy concerns.
- **Flexible work arrangements:** Implement policies to accommodate diverse work schedules and locations.

4.1.2 Job analysis, job specification, job description

In the context of ICT-based projects and organizations, "job analysis" is the process of systematically studying a job to identify its key tasks, responsibilities, required skills, and necessary qualifications, while a "job description" outlines the duties and responsibilities of a specific role, and a "job specification" details the specific skills, experience, and qualifications needed to perform that job effectively; essentially, the job analysis provides the foundation for creating both the job description and job specification.

Breakdown:

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- **Job Analysis:**
  - **Purpose:** To thoroughly understand the demands of a position within an ICT project or organization, including technical skills, problem-solving abilities, communication requirements, and project management skills needed.
  - **Methods:** Interviews with current employees, observation of work activities, reviewing project documentation, analyzing industry standards.

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- **Key aspects considered:**
    - Specific software proficiency (e.g., coding languages like Python, Java, databases like MySQL)
    - Hardware knowledge (network infrastructure, servers)
    - Understanding of project methodologies (Agile, Waterfall)
    - Ability to work in a collaborative environment

- **Job Description:**
    - **Purpose:** A clear and concise overview of the job responsibilities, reporting structure, and key performance indicators for a specific ICT role.
    - **Example elements:**
        - Develop and maintain web applications using front-end technologies (HTML, CSS, JavaScript)
        - Troubleshoot network connectivity issues
        - Collaborate with cross-functional teams to deliver projects on time
- **Job Specification:**
    - **Purpose:** A detailed list of qualifications, skills, and experience required to successfully perform the job outlined in the job description.
    - **Example elements:**
        - Bachelor's degree in Computer Science or related field
        - 3+ years of experience in software development
        - Proficiency in cloud computing platforms (AWS, Azure)
        - Excellent communication and problem-solving skills

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How they relate in ICT projects:

- **Recruitment:**

  The information gathered from a job analysis is crucial for crafting targeted job descriptions and specifications, allowing for efficient recruitment of qualified candidates for ICT roles.

- **Performance Management:**

  By clearly defining job requirements, job descriptions and specifications can serve as a basis for employee performance evaluations.

- **Training and Development:**
  Identifying skill gaps through job analysis can inform the design of targeted training programs to upskill existing ICT staff.

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4.1.3 Recruitment and selection



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In Human Resource Management (HRM), the recruitment and selection process refers to a systematic approach to identifying, attracting, evaluating, and hiring qualified candidates to fill open positions within an organization, typically involving steps like job analysis, job posting, candidate sourcing, screening, interviewing, testing, and making a final hiring decision.

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Key Stages of the Recruitment and Selection Process:

- **Job Analysis:**
  - Identifying the specific responsibilities, duties, and required skills for a vacant position through a thorough job analysis.
  - This forms the basis for creating a detailed job description.
- **Planning and Forecasting:**
  - Determining staffing needs based on organizational goals and future projections.
  - Identifying potential vacancies and prioritizing hiring requirements.
- **Job Posting and Advertising:**
  - Creating a compelling job description and posting it on relevant job boards, company websites, internal platforms, or through other recruitment channels.
- **Candidate Sourcing:**
  - Actively seeking out potential candidates through various methods like employee referrals, headhunting, social media, networking, and job fairs.
- **Initial Screening:**
  - Reviewing resumes and applications to shortlist candidates based on their qualifications and fit with the job requirements.
- **Preliminary Interviews:**
  - Conducting phone or video interviews with shortlisted candidates to further assess their basic skills, experience, and interest in the role.
- **Employment Testing:**
  - Administering relevant skills tests, aptitude tests, personality assessments, or other assessments to evaluate candidates' abilities in specific areas.
- **In-Depth Interviews:**
  - Conducting comprehensive face-to-face interviews with shortlisted candidates to delve deeper into their experience, qualifications, and behavioral fit for the role.
- **Reference Checks:**

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    - Contacting previous employers or professional references provided by candidates to verify their employment history and skills.
- **Decision Making and Job Offer:**
    - Selecting the most qualified candidate based on the evaluation process and extending a formal job offer.
- **Onboarding:**
    - A structured process to introduce the new employee to the organization, their role, and company culture.

Important Considerations in Recruitment and Selection:

- **Legal Compliance:**

  Ensuring adherence to employment laws and regulations regarding non-discrimination, equal opportunity, and privacy.

- **Diversity and Inclusion:**

  Actively seeking a diverse pool of candidates to promote inclusivity within the organization.

- **Candidate Experience:**

  Providing a positive experience for all applicants throughout the recruitment process.

- **Cost-Effectiveness:**
  Optimizing the recruitment process to minimize costs while maintaining quality standards.

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4.1.4 Human resource training (on the job and off the job)

- Training plays a crucial role in the development of employees, enabling them to enhance their knowledge and skills to drive improved performance and capabilities in the workplace. When it comes to employee training, businesses have a range of options available to them.
- **On-the-job training** offers hands-on experience within the workplace, fostering the immediate application of skills and integration with company culture. Conversely, off-the-job training provides structured programs and theoretical knowledge, suitable for broader skill development.



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Apprenticeship Programs: Apprenticeship programs place the trainee under the guidance of well-trained personnel. These programs are designed to obtain skills and knowledge of higher levels. Such programs are necessary for people entering skilled jobs, like, plumbers, electricians, IT technician etc.

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Off-the-Job Training

- Off-the-job training involves employees receiving training away from their regular workspace and job roles.
- This type of training can occur offsite or in dedicated training rooms within the organization.
- Off-the-job training often takes a more formal and theoretical approach compared to on-the-job training.

Common methods of off-the-job training include:

- **Computer-Based Training**: Employees undertake training using computer-based software or e-learning modules. This method is commonly used for initial employee induction or for annual refresher training on topics such as safety procedures.
- **Outside Trainers**: Specialists or experts from external organisations deliver training to enhance the learning process. These trainers possess in-depth knowledge and experience, often surpassing what internal resources can provide. They can deliver training on-site or off-site.
- **Sandwich Courses**: Employees spend an extended period away from the workplace to focus solely on their training. This can involve studying qualifications at local colleges or universities before returning to work equipped with relevant knowledge and skills.

Off-the-job training offers advantages such as a distraction-free environment, expert trainers, and opportunities for collaboration and networking with other trainees.

However, it may not directly align with the equipment, machinery, or software used in the workplace, necessitating on-the-job training afterward. Additionally, off-the-job training often incurs higher costs due to external training providers and associated expenses like travel and accommodation.

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4.1.5 Performance appraisal and methods

- Performance appraisal is a process of documenting and evaluating the performance of an employee.
- It's also sometimes referred to as performance review, performance evaluation, or employee appraisal. Different organizations adopt different types of performance appraisal methods based on their needs and goals.
- A performance appraisal is a process that evaluates an employee's job performance and contributions to a company. It's also known as an annual review, employee appraisal, performance review, or evaluation.

What's it used for?

- Identify strengths and weaknesses
- Determine areas for improvement
- Help with career growth
- Improve performance
- Increase employee engagement
- Help determine training
- Clarify expectations
- Evaluate goals
- Provide documentation

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How's it done?

(Methods)

- Managers and employees set goals together
- Employees evaluate their own performance
- Managers and employees discuss progress
- Managers evaluate whether goals were achieved
- Managers and human resources staff use an appraisal form to record ratings and comments

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What are some methods?

- **Management by objectives (MBO)**: A process where managers and employees set clear goals together
- **Self-assessment**: Employees answer questions about their job role to evaluate their performance

Benefits

- Promotes regular communication and teamwork
- Keeps evaluations fair and focused on clear results

4.1.6 Challenges in managing people in ICT workforce

Managing people in an ICT workforce presents challenges like rapid technological change, maintaining employee engagement in a fast-paced environment, ensuring diverse skillsets are developed, managing remote work, addressing potential burnout from high pressure, navigating cultural differences in global teams, and constantly adapting to new tools and technologies to stay competitive.

Key challenges include:

- **Rapid Skill Obsolescence:**

  The ICT field evolves quickly, meaning employees need continuous learning to stay updated with the latest technologies, which can be a challenge for managers to facilitate.

- **Attracting and Retaining Talent:**

  Finding qualified candidates with the necessary skills and experience in a competitive market, while also providing opportunities for career growth to retain top talent.

- **Work-Life Balance:**

  The demanding nature of ICT jobs, with potential for long hours and on-call shifts, can lead to burnout and issues with work-life balance.

- **Managing Remote Work:**

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Effectively supervising and maintaining team cohesion when employees work remotely, ensuring clear communication and collaboration across different time zones.

- **Diversity and Inclusion:**

  Creating a diverse and inclusive workplace, where individuals from different backgrounds feel valued and respected.

- **Change Management:**

  Implementing new technologies and processes smoothly, addressing resistance to change within the team.

- **Communication Challenges:**

  Maintaining clear and open communication channels, especially when managing geographically dispersed teams.

- **Performance Measurement:**

  Defining clear and relevant performance metrics for roles that involve rapidly evolving technologies.

- **Cybersecurity Concerns:**

  Ensuring employees are aware of and adhere to cybersecurity protocols to protect sensitive data.

- **Employee Engagement:**
  Fostering a positive work environment and motivating employees to stay engaged in their roles, especially when faced with constant change.

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Strategies to address these challenges:

- **Invest in Training and Development:**

  Provide ongoing training programs to keep employees up-to-date with the latest technologies and skills.

- **Flexible Work Arrangements:**

  Offer flexible work schedules and remote work options to improve work-life balance.

- **Clear Career Paths:**

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Establish clear career progression opportunities to motivate employees and encourage retention.

- **Open Communication Channels:**

  Encourage open communication and regular feedback loops between managers and employees.

- **Employee Wellness Initiatives:**

  Implement programs to promote mental and physical well-being.

- **Strong Leadership:**

  Develop strong leadership skills to guide teams through change and provide support.

- **Diversity and Inclusion Practices:**
  Implement initiatives to promote diversity and inclusion within the workforce.

4.2 Control

In Human Resource Management (HRM), the "control process" refers to a systematic method of monitoring employee performance against pre-established standards, identifying deviations from those standards, and taking corrective actions to ensure alignment with organizational goals and objectives; essentially, it involves setting performance expectations, measuring actual performance, comparing it to the standards, and taking necessary adjustments when needed.

4.2.1 Importance

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Importance of the control process in HRM:

- **Ensures alignment with organizational goals:**

  By monitoring employee performance, HRM can identify areas where adjustments are needed to achieve organizational objectives.

- **Improves employee performance:**

  Providing feedback and targeted interventions can help employees improve their skills and performance levels.

- **Identifies training needs:**

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Analyzing performance data can reveal areas where employees require additional training or development opportunities.

- **Motivates employees:**
  Regular feedback and recognition of achievements can foster a sense of accountability and motivation among employees.

4.2.2 Process and types

Key steps in the HRM control process:

- **Establishing Standards:**

  Defining clear performance expectations for employees, including measurable criteria like productivity targets, quality standards, attendance rates, and behavioral competencies.

- **Measuring Performance:**

  Gathering data on employee performance through methods like performance appraisals, employee feedback surveys, productivity reports, and attendance records.

- **Comparing Performance to Standards:**

  Analyzing the collected data to identify areas where employee performance deviates from the established standards.

- **Identifying Causes of Deviation:**

  Investigating the reasons behind performance gaps, which could include factors like lack of training, inadequate resources, motivational issues, or external challenges.

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- **Taking Corrective Action:**
  Implementing appropriate actions to address performance issues, which may include providing additional training, coaching, performance improvement plans, addressing employee concerns, or making necessary adjustments to work processes.

4.2.3 Techniques

Types of control mechanisms in HRM:

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- **Performance Appraisals:**

  Regular evaluations of employee performance against pre-set goals, often involving feedback from supervisors and peers.

- **Employee Monitoring Systems:**

  Tracking employee activity and productivity through digital tools, ensuring compliance with work standards.

- **Attendance Tracking:**

  Monitoring employee attendance to identify patterns of absenteeism and address potential issues.

- **Employee Feedback Surveys:**

  Gathering employee opinions and perceptions regarding work environment, job satisfaction, and management practices to identify areas for improvement.

- **Quality Control Checks:**
  Assessing the quality of work produced by employees to ensure consistency with standards.

4.2.4 ICT tools for effective control of engineering projects and organizations.

ICT tools for effective control of engineering projects and organizations include:

- project management software like **Microsoft Project, Jira, Wrike, ClickUp, and Accelo**
- ***These tools facilitate task allocation, resource management, scheduling and planning, automated reporting, and streamlined communication, enabling better oversight and control over complex engineering projects.***

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Specific examples of ICT tools for engineering project control:

- **Microsoft Project:**

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Comprehensive project management tool with advanced scheduling capabilities, resource management, and reporting features.

- **Jira:**

  Popular for agile project management, particularly suited for software development projects, with robust issue tracking and reporting functions.

- **Wrike:**

  Strong emphasis on task dependencies and visual workflows, helpful for complex engineering projects with intricate relationships between tasks.

- **ClickUp:**

  Versatile project management platform offering a wide range of features including task management, time tracking, and custom workflows.

- **Accelo:**

  Accelo is a cloud-based business management platform that helps professional services businesses manage their client work. It includes modules for sales, projects, ticketing, time tracking, billing, and reporting.

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Key features of these tools for engineering project control:

- **Task management:**

  Assign tasks to team members, set deadlines, track progress, and visualize task dependencies.

- **Resource allocation:**

  Manage team capacity, allocate resources efficiently across projects, and monitor resource utilization.

- **Gantt charts:**

  Visualize project timelines, milestones, and critical paths for effective planning and monitoring.

- **Collaboration features:**

  Facilitate communication through shared document access, comments, and real-time updates.

- **Reporting and dashboards:**
  Generate automated reports on project performance metrics like budget, schedule, and quality, providing insights for decision-making.

Important considerations when selecting ICT tools for engineering project control:

- **Project complexity:**

  Choose a tool with features that align with the scale and intricacy of your projects.

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- **Team size:**

  Consider the number of users and their needs for collaboration and communication.

- **Integration capabilities:**

  Ensure the tool can integrate with other systems used by your organization, like CRM or CAD software.

- **User interface:**
  Select a tool with an intuitive design that is easy for your team to learn and use.

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